Komalpreet Kaur
Gola Gokarannath, Uttar Pradesh 262802
+91 9451735039 kaurkomalpreetsohal@gmail.com linkedin.com/in/komalpreetkaur-k/ github.com/Komalpreet2809

Projects

Vanta | *PyTorch, SpeechBrain, FastAPI, Next.js, Docker, Hugging Face* **Mar 2026 - Apr 2026**

- Built a Target Speaker Extraction system isolating a target voice from noisy recordings using a 5-second reference clip.
- Trained Conv-TasNet from scratch with SI-SDR loss; paired with frozen ECAPA-TDNN encoder for cosine-similarity-based voice conditioning.
- Deployed FastAPI backend on Hugging Face Spaces and Next.js frontend on Vercel.
- Repository Link: **https://github.com/Komalpreet2809/Vanta**

Soma | *FastAPI, React, LangChain, Neo4j, ChromaDB, SQLite, Docker* **Mar 2026 - Apr 2026**

- Designed a brain-inspired AI system with multi-layered memory architecture for context-aware reasoning.
- Implemented hybrid RAG pipeline combining ChromaDB and Neo4j with LangGraph for multi-step AI decision-making.
- Built full-stack system with FastAPI and React for consolidation and knowledge refinement.
- Repository *Link:* **https://github.com/Komalpreet2809/SOMA**

Specula | *Python, FastAPI, PyTorch, HuggingFace, OpenCV* **Feb 2026 - Mar 2026**

- Developed a full-stack AI tool using a 5-method forensic pipeline for deepfake image detection.
- Implemented CNN classification with Grad-CAM explainability and analyzers like ELA, FFT, and metadata forensics.
- Designed a weighted scoring system to generate confidence-based authenticity verdicts and enabled batch + PDF reporting APIs.
- Repository Link: **https://github.com/Komalpreet2809/Specula**

Qlothi | *Python, FastAPI, PyTorch, HuggingFace, OpenCV, JavaScript* **Jan 2026 - Mar 2026**

- Built a Chrome extension for Pinterest enabling AI-based outfit segmentation and product discovery.
- Integrated SegFormer model for clothing detection and automated visual search using Playwright + Google Lens.
- Developed FastAPI backend and interactive UI with filtering features for style, budget, and category-based recommendations.
- Repository Link: **https://github.com/Komalpreet2809/Qlothi**

Certifications

Social Networks **Jan2025 - Apr 2025**
NPTEL — Certificate Link

Skills

Languages: Python, Java, C++, Bash
Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning, Computer Vision, NLP, Feature Engineering
Generative AI & RAG: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV, NumPy, Pandas
Backend & APIs: FastAPI, REST APIs, Model Deployment
Frontend & Tools: React, JavaScript, HTML, CSS, Chrome Extensions
Databases: ChromaDB (Vector DB), Neo4j (Graph DB), SQL (SQLite), NoSQL
MLOps & Systems: Docker, Git, GitHub Actions, CI/CD, Linux
Cloud & Platforms: AWS, Azure

Education

Lovely Professional University **2023 – 2027**
Bachelor of Technology - Computer Science and Engineering — CGPA: 8.56 *Phagwara, Punjab*

Shri Rajendra Giri Memorial Academy **2021 – 2022**
12th — Percentage: 93% *Kheri, Uttar Pradesh*

Shri Guru Nanak Dev Public School **2019 – 2020**
10th — Percentage: 96.2% *Kheri, Uttar Pradesh*